URSODEOXYCHOLIC ACID

Cholestatic liver disease

Take with food

CHEMICAL NAME 3α,7β-dihydroxy-5β-cholan-24-oic acid (ursodiol)
CLASS Hydrophilic bile acid
TRADE NAMES Ursofalk · Destolit · Urso · Actigall

THERAPEUTIC USES / INDICATIONS

- Primary biliary cholangitis (PBC); first-line.
- Dissolution of small, radiolucent cholesterol gallstones (functioning gallbladder).
- Intrahepatic cholestasis of pregnancy (itch).
- Cholestasis in cystic fibrosis, TPN; prevention of gallstones in rapid weight loss.

ADMINISTRATION

- **PO** with food (or evening meal for gallstones); take regularly.
- Separate from antacids and bile-acid sequestrants by 2 h.

DOSAGES

- **PBC:** 13–15 mg/kg/day in 2–4 divided doses.
- **Gallstone dissolution:** 8–12 mg/kg/day (single bedtime dose or divided) for up to 2 years.
- **Cholestasis of pregnancy:** 500 mg–2 g/day in divided doses.
- **Children (CF cholestasis):** 20 mg/kg/day.

PHARMACOKINETICS

- Absorbed in small intestine; conjugated in liver; enterohepatic circulation.
- Excreted in faeces.

PHARMACODYNAMICS

- Replaces toxic hydrophobic bile acids → cytoprotective to hepatocytes and bile ducts.
- ↓ cholesterol saturation of bile → dissolves cholesterol stones; stimulates bile flow; immunomodulatory.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, loose stools.
- Abdominal pain (right upper quadrant).
- Pruritus (transient).

ADVERSE EFFECTS (SERIOUS)

- Calcification of gallstones.
- Decompensation of advanced cirrhosis (rare).
- Urticaria.

CONTRAINDICATIONS & CAUTIONS

- Acute inflammation of gallbladder/bile ducts; biliary obstruction.
- Non-functioning gallbladder; calcified or radio-opaque stones; frequent biliary colic.
- Advanced decompensated cirrhosis (relative).

KEY INTERACTIONS

- **Cholestyramine, colestipol, aluminium antacids:** ↓ absorption.
- **Ciclosporin:** ↑ absorption — monitor levels.
- **Oestrogens, fibrates:** ↑ biliary cholesterol (reduce efficacy).

NURSING MANAGEMENT

- Monitor LFTs (ALP, bilirubin) every 3–6 months.
- Gallstones: ultrasound at 6–12 months.
- Assess pruritus, jaundice.
- Pregnancy: monitor bile acids and fetal wellbeing.

PATIENT EDUCATION

- Take with meals regularly; long-term treatment.
- Report diarrhoea or abdominal pain.
- Avoid antacids around the dose.

KEY POINTS

- Hydrophilic bile acid — cytoprotective.
- Dissolves small cholesterol stones.
- First-line for PBC (13–15 mg/kg/day).
- Cholestasis of pregnancy itch.